Graphic Era HILL UNIVERSITY

End Term (Odd) Semester Examination December 2024

Roll no.....

Name of the Course and semester: B. Tech. C.E. and 7th sem
Name of the Paper: *Ground Improvement Techniques*
Paper Code: TCE-722
Time: 3 hours Maximum Marks: 100

Note:

(i) All the questions are compulsory.
(ii) Answer any two sub questions from a, b and c in each main question.
(iii) Total marks for each question is 20 (twenty).
(iv) Each sub-question carries 10 marks.

Q1. (2X10=20 Marks) CO1

a. Assess the role of lime stabilization in enhancing the strength and stability of lateritic soil.
b. Compare the effectiveness of sand drains and wick drains in improving soft alluvial soil deposits.
c. Define ground improvement and explain its importance in foundation engineering.

Q2. (2X10=20 Marks) CO2

a. Propose a suitable method for densifying loose granular soil at a depth of 10 m for a construction project.
b. Discuss the advantages and limitations of preloading as a ground improvement technique for cohesive soils.
c. Discuss the feasibility of thermal methods for densifying a highly plastic clay soil.

Q3. (2X10=20 Marks) CO3

a. Describe the field control measures necessary during the application of bituminous stabilization in road construction.
b. Explain the mechanisms behind cement stabilization and its impact on soil properties.
c. Explain the differences between ascending and descending grouting methods, highlighting their applications.

Q4. (2X10=20 Marks) CO4

a. Discuss the mechanisms of swelling in expansive soils and describe how these mechanisms can affect structures built on them.
b. Evaluate the effectiveness of soil stabilization methods (lime, cement, and chemical treatments) in improving expansive soils.
c. Explain the types of geo-textiles and their primary functions in geotechnical engineering. Discuss their application in slope stability and drainage.

Q5. (2X10=20 Marks) CO5

a. Define reinforced earth and explain its fundamental principles. List the main components of a reinforced earth structure.
b. Propose a reinforced earth wall solution for a highway passing through a hilly region. Highlight the design steps, construction challenges, and measures to ensure long-term stability.
c. Assess the advantages and limitations of using reinforced earth walls compared to conventional retaining walls in urban infrastructure projects.